

V-SPACE FPGA BUILD CHALLENGE 2026

# OVERALL PROJECT DOCUMENTATION

Intel DE10-Nano FPGA Development Board

Five RTL design demonstrations: safety monitoring, energy scaling, machine health, traffic monitoring, and conveyor inspection.

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Platform: Intel DE10-Nano • FPGA: Cyclone V SoC

This document combines project objectives, design logic, synthesizable RTL, test cases, hardware evidence, applications, and conclusions.

# Contents and Submission Structure

Each project is documented using the requested seven sections: (1) Objective, (2) Block Diagram, (3) RTL Design, (4) Simulation Results, (5) Hardware Implementation, (6) Applications, and (7) Conclusion. Simulation tables below describe expected results from the provided RTL and test cases; screenshots/transcripts should be retained in each project's Simulation folder.

## Board interface notes

The DE10-Nano switches are used as 4-bit inputs where applicable. The onboard LEDs are active-high outputs in these designs. Switch positions and displayed values must be checked against the exact pin assignments in the Quartus project before final submission.

## Important engineering note

These are educational combinational-logic prototypes. The energy project is not a calibrated electrical meter; the traffic project is not a certified traffic controller; the health and conveyor projects use simplified sensor encodings and would need real sensors, conditioning, timing, and safety validation for deployment.

## Common hardware pin reference

| Board signal | FPGA pin |
|--------------|----------|
| SW0          | Y24      |
| SW1          | W24      |
| SW2          | W21      |
| SW3          | W20      |
| LED0         | W15      |
| LED1         | AA24     |
| LED2         | V16      |
| LED3         | V15      |
| LED4         | AF26     |
| LED5         | AE26     |
| LED6         | Y16      |
| LED7         | AA23     |

# Project 1 — Beginner

## Smart Motion and Accident Detection

### 1. Objective

Demonstrate a simple FPGA-based alert logic in which motion and impact inputs are monitored and an accident indicator activates only when impact and tilt are both asserted.

### 2. Block Diagram

**Inputs:** motion, impact, tilt (1-bit inputs)

**Processing:** motion\_led mirrors motion; impact\_led mirrors impact; accident\_led = impact AND tilt.

**Outputs:** motion\_led, impact\_led, accident\_led (1-bit outputs)

### 3. RTL Design

**Inputs:** motion, impact, tilt (1-bit inputs)

**Outputs:** motion\_led, impact\_led, accident\_led (1-bit outputs)

**Logic summary:** motion\_led mirrors motion; impact\_led mirrors impact; accident\_led = impact AND tilt.

```
module Smart_Accident_Detection (
    input  wire motion,
    input  wire impact,
    input  wire tilt,
    output wire motion_led,
    output wire impact_led,
    output wire accident_led
);
    assign motion_led = motion;
    assign impact_led = impact;
    assign accident_led = impact & tilt;
endmodule
```

### 4. Simulation Results

| Input case | Expected outputs | Interpretation                                        |
|------------|------------------|-------------------------------------------------------|
| 000        | 000              | All outputs LOW                                       |
| 100        | 100              | Motion LED HIGH                                       |
| 010        | 010              | Impact LED HIGH; accident LED LOW                     |
| 011        | 011              | Impact and tilt HIGH; accident LED HIGH               |
| 111        | 111              | All three indicators HIGH                             |
| 110        | 110              | Impact LED HIGH; accident LED LOW because tilt is LOW |

Testbench timing for Project 1 uses 10 ns per input case; the six cases run from 0 to 60 ns. A final \$stop at 60 ns is expected.

### 5. Hardware Implementation

The RTL is intended to be compiled for the DE10-Nano Cyclone V FPGA and assigned to onboard switches and LEDs using the project pin assignments. The photos below document the powered board state supplied for this project. LED patterns in photos can be affected by switch positions, camera exposure, and viewing orientation.

Pin assignment summary

| Signal     | Assignment  |
|------------|-------------|
| motion     | SW0 / Y24   |
| impact     | SW1 / W24   |
| tilt       | SW2 / W21   |
| motion_led | LED0 / W15  |
| impact_led | LED1 / AA24 |

| Signal       | Assignment |
|--------------|------------|
| accident_led | LED2 / V16 |

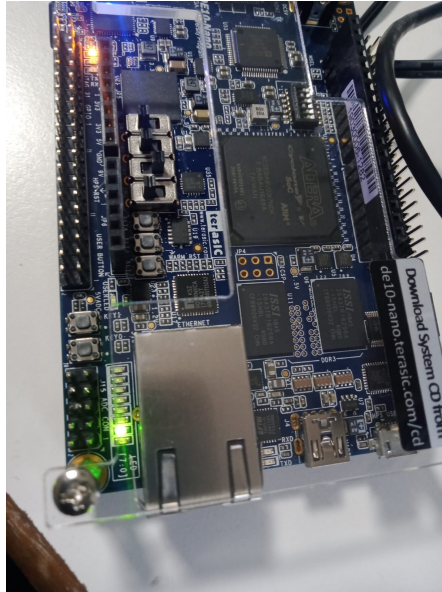
## 6. Applications

Prototype vehicle-impact alerts, motion-triggered monitoring, and educational demonstrations of combinational safety logic.

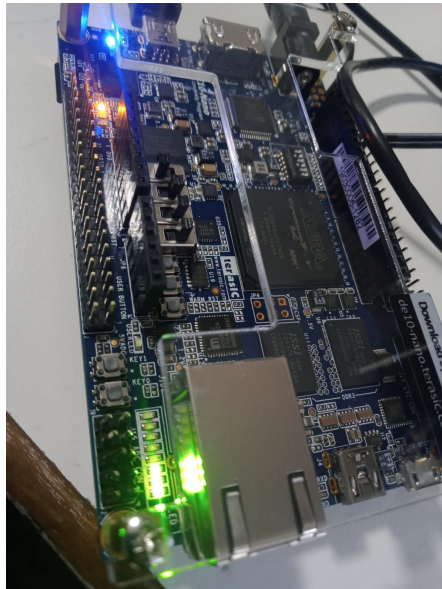
## 7. Conclusion

The Smart Motion and Accident Detection project demonstrates a compact digital-logic design mapped to an FPGA. The expected output table provides a reference for checking the RTL in simulation and on the board. For a complete submission, include the actual Questa waveform, transcript, Quartus compilation evidence, and video link in the corresponding repository folders.

## Project 1 — Beginner — Hardware Photo Evidence



Smart Motion and Accident Detection hardware evidence — photo 1



Smart Motion and Accident Detection hardware evidence — photo 2

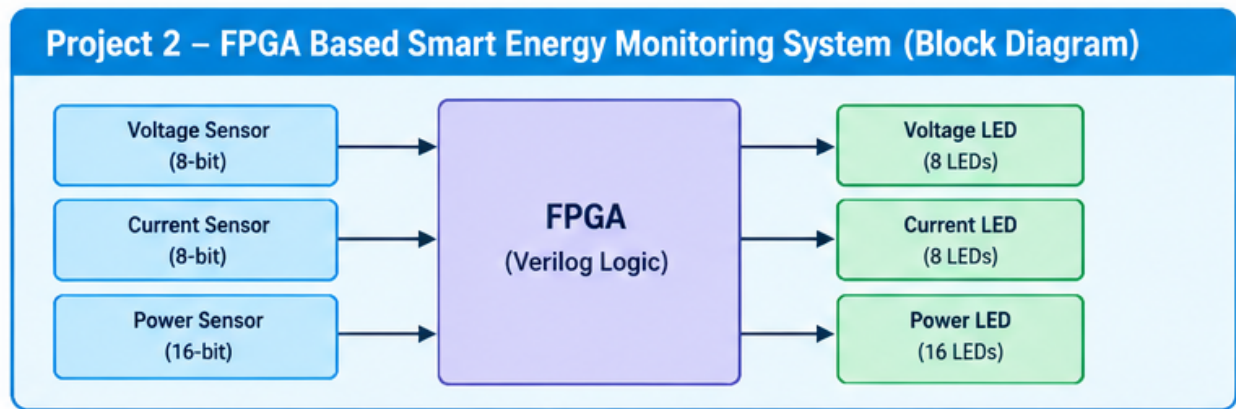
## Project 2 — Beginner

### FPGA-Based Smart Energy Monitoring System

#### 1. Objective

Convert a 4-bit input value into a scaled power indication and raise an overload flag when the calculated value exceeds 100.

#### 2. Block Diagram



FPGA-Based Smart Energy Monitoring System — block diagram

#### 3. RTL Design

**Inputs:** input\_value[3:0] (4-bit unsigned value)

**Outputs:** power[6:0] (7-bit displayed value), overload (1-bit flag)

**Logic summary:** calculated\_power = input\_value × 10. power displays the low 7 bits of the calculated result; overload is HIGH when calculated\_power > 100.

```
module Smart_Energy_Monitoring (
    input wire [3:0] input_value,
    output wire [6:0] power,
    output wire overload
);
    wire [7:0] calculated_power;
    assign calculated_power = input_value * 8'd10;
    assign power = calculated_power[6:0];
    assign overload = (calculated_power > 8'd100);
endmodule
```

#### 4. Simulation Results

|  | Expected outputs | Interpretation |                                                       |
|--|------------------|----------------|-------------------------------------------------------|
|  | 0                | 0              | No overload                                           |
|  | 50               | 0              | No overload                                           |
|  | 80               | 0              | No overload                                           |
|  | 100              | 0              | Boundary value; no overload                           |
|  | 110              | 1              | Overload asserted                                     |
|  | 22*              | 1              | Calculated power is 150; 7-bit output wraps to 150 mo |

\*The 7-bit power output stores only bits [6:0], so values above 127 wrap around. The overload comparison uses the full 8-bit calculated\_power and remains correct.

#### 5. Hardware Implementation

The RTL is intended to be compiled for the DE10-Nano Cyclone V FPGA and assigned to onboard switches and LEDs using the project pin assignments. The photos below document the powered board state supplied for this project. LED patterns in photos can be affected by switch positions, camera exposure, and viewing orientation.

Pin assignment summary

| Signal            | Assignment                                        |
|-------------------|---------------------------------------------------|
| input_value[0..3] | SW0..SW3 / Y24, W24, W21, W20                     |
| power[0..6]       | LED0..LED6 / W15, AA24, V16, V15, AF26, AE26, Y16 |
| overload          | LED7 / AA23                                       |

## 6. Applications

Learning prototype for threshold monitoring, digital power scaling, and overload indication. It is not a calibrated mains-energy meter.

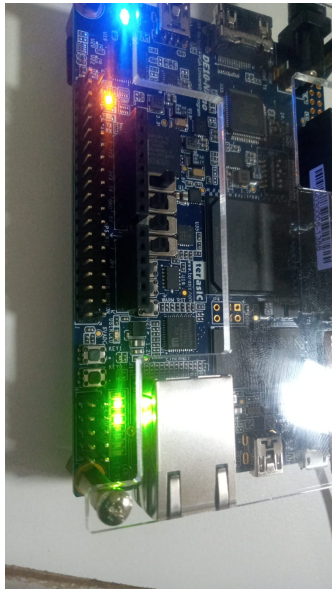
## 7. Conclusion

The FPGA-Based Smart Energy Monitoring System project demonstrates a compact digital-logic design mapped to an FPGA. The expected output table provides a reference for checking the RTL in simulation and on the board. For a complete submission, include the actual Questa waveform, transcript, Quartus compilation evidence, and video link in the corresponding repository folders.

## Project 2 — Beginner — Hardware Photo Evidence



FPGA-Based Smart Energy Monitoring System hardware evidence — photo 1



FPGA-Based Smart Energy Monitoring System hardware evidence — photo 2



FPGA-Based Smart Energy Monitoring System hardware evidence — photo 3

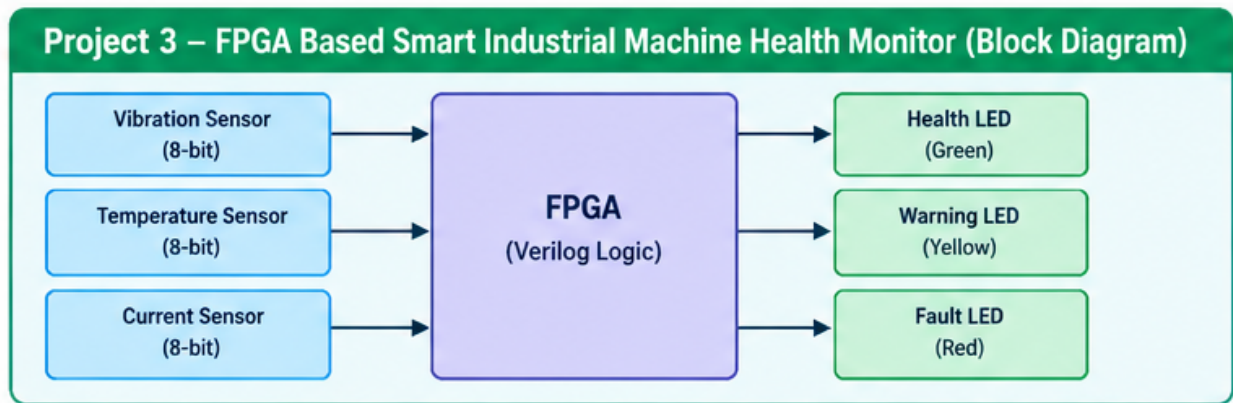
# Project 3 — Intermediate

## Industrial Machine Health Monitoring System

### 1. Objective

Classify a 4-bit sensor reading into normal, warning, and critical ranges, with an alarm output for critical readings.

### 2. Block Diagram



Industrial Machine Health Monitoring System — block diagram

### 3. RTL Design

**Inputs:** sensor\_value[3:0] (4-bit unsigned sensor reading)

**Outputs:** normal\_led, warning\_led, critical\_led, alarm

**Logic summary:** Normal for 0–7; warning for 8–11; critical and alarm for 12–15.

```
module Industrial_Machine_Health (
    input wire [3:0] sensor_value,
    output wire normal_led,
    output wire warning_led,
    output wire critical_led,
    output wire alarm
);
    assign normal_led = (sensor_value <= 4'd7);
    assign warning_led = (sensor_value >= 4'd8) && (sensor_value <= 4'd11);
    assign critical_led = (sensor_value >= 4'd12);
    assign alarm = (sensor_value >= 4'd12);
endmodule
```

### 4. Simulation Results

| Input case | Expected outputs    | Interpretation                 |
|------------|---------------------|--------------------------------|
| 0, 7       | Normal=1            | Warning=0, Critical=0, Alarm=0 |
| 8, 10, 11  | Warning=1           | Normal=0, Critical=0, Alarm=0  |
| 12, 15     | Critical=1, Alarm=1 | Normal=0, Warning=0            |

### 5. Hardware Implementation

The RTL is intended to be compiled for the DE10-Nano Cyclone V FPGA and assigned to onboard switches and LEDs using the project pin assignments. The photos below document the powered board state supplied for this project. LED patterns in photos can be affected by switch positions, camera exposure, and viewing orientation.

Pin assignment summary

| Signal             | Assignment                    |
|--------------------|-------------------------------|
| sensor_value[0..3] | SW0..SW3 / Y24, W24, W21, W20 |
| normal_led         | LED0 / W15                    |
| warning_led        | LED1 / AA24                   |

| Signal       | Assignment |
|--------------|------------|
| critical_led | LED2 / V16 |
| alarm        | LED3 / V15 |

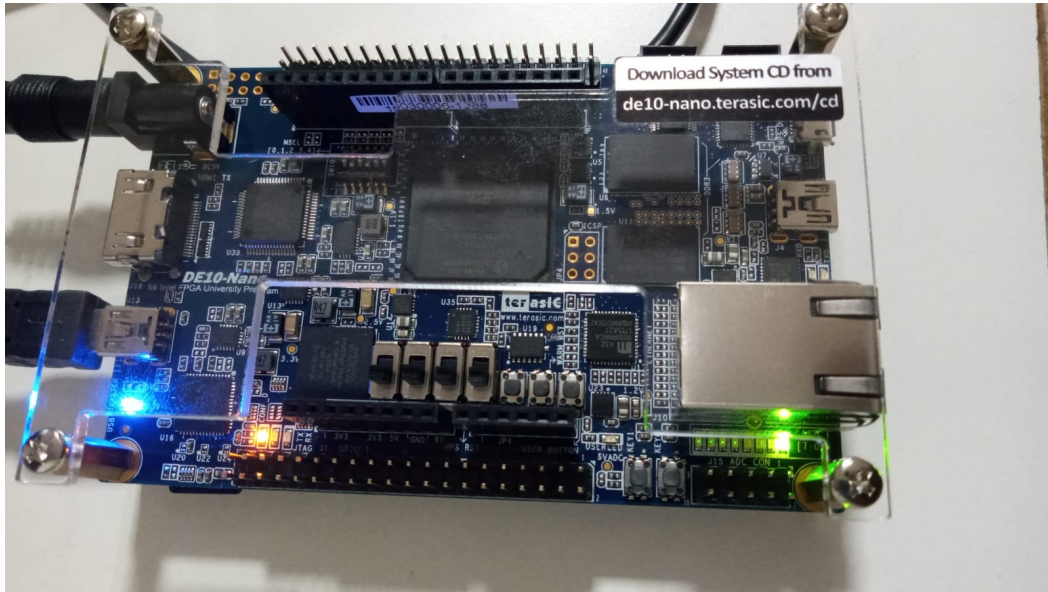
## 6. Applications

Demonstration of threshold-based condition monitoring for motors, pumps, temperature-like readings, and industrial training systems.

## 7. Conclusion

The Industrial Machine Health Monitoring System project demonstrates a compact digital-logic design mapped to an FPGA. The expected output table provides a reference for checking the RTL in simulation and on the board. For a complete submission, include the actual Questa waveform, transcript, Quartus compilation evidence, and video link in the corresponding repository folders.

## Project 3 — Intermediate — Hardware Photo Evidence



Industrial Machine Health Monitoring System hardware evidence — photo 1



Industrial Machine Health Monitoring System hardware evidence — photo 2

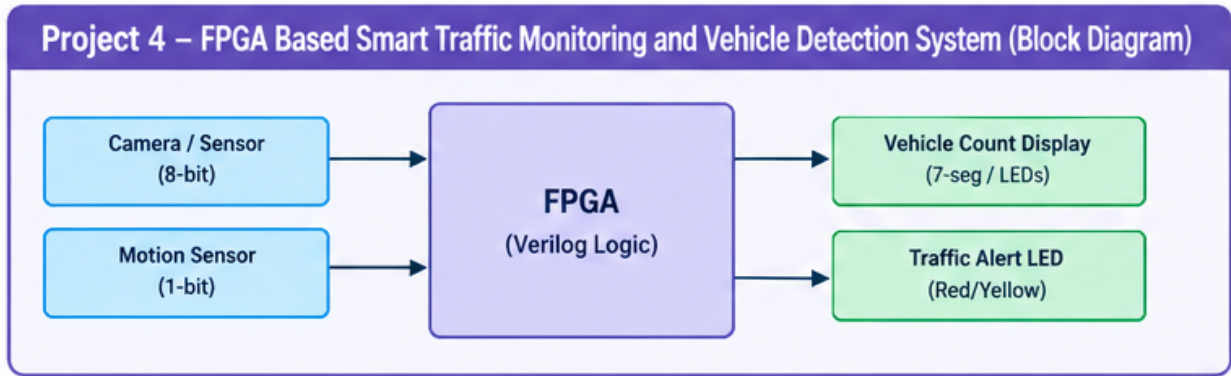
# Project 4 — Intermediate

## FPGA-Based Traffic Monitoring and Signal Control System

### 1. Objective

Use a 4-bit traffic-sensor pattern to choose a simple signal state and flag heavy traffic when at least two sensor bits are HIGH.

### 2. Block Diagram



FPGA-Based Traffic Monitoring and Signal Control System — block diagram

### 3. RTL Design

**Inputs:** traffic\_sensor[3:0]

**Outputs:** red\_led, yellow\_led, green\_led, heavy\_traffic

**Logic summary:** Heavy traffic is true when any pair of sensor bits is simultaneously HIGH. Red is asserted for heavy traffic; yellow for nonzero traffic that is not heavy; green when all sensor bits are zero.

```
module Traffic_Monitoring (
    input wire [3:0] traffic_sensor,
    output wire red_led,
    output wire yellow_led,
    output wire green_led,
    output wire heavy_traffic
);
    assign heavy_traffic =
        (traffic_sensor[0] & traffic_sensor[1]) |
        (traffic_sensor[0] & traffic_sensor[2]) |
        (traffic_sensor[0] & traffic_sensor[3]) |
        (traffic_sensor[1] & traffic_sensor[2]) |
        (traffic_sensor[1] & traffic_sensor[3]) |
        (traffic_sensor[2] & traffic_sensor[3]);
    assign red_led = heavy_traffic;
    assign yellow_led = ~heavy_traffic & (~|traffic_sensor);
    assign green_led = ~(|traffic_sensor);
endmodule
```

### 4. Simulation Results

| Input case | Expected outputs | Interpretation           |
|------------|------------------|--------------------------|
| 0000       | Green=1          | Yellow=0, Red=0, Heavy=0 |
| 0001       | Yellow=1         | Green=0, Red=0, Heavy=0  |
| 0011       | Red=1, Heavy=1   | Green=0, Yellow=0        |
| 1111       | Red=1, Heavy=1   | Green=0, Yellow=0        |

### 5. Hardware Implementation

The RTL is intended to be compiled for the DE10-Nano Cyclone V FPGA and assigned to onboard switches and LEDs using the project pin assignments. The photos below document the powered board state supplied for this project. LED patterns in photos can be affected by switch positions, camera exposure, and viewing orientation.

Pin assignment summary

| Signal               | Assignment                    |
|----------------------|-------------------------------|
| traffic_sensor[0..3] | SW0..SW3 / Y24, W24, W21, W20 |
| red_led              | LED0 / W15                    |
| yellow_led           | LED1 / AA24                   |
| green_led            | LED2 / V16                    |
| heavy_traffic        | LED3 / V15                    |

## 6. Applications

Traffic-density demonstration, basic signal-state logic, and educational prototyping. A real traffic controller requires timing, fail-safe states, and field validation.

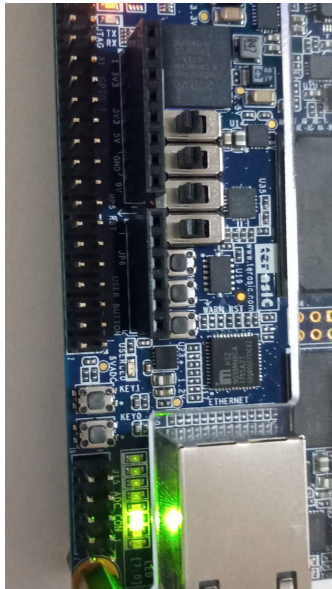
## 7. Conclusion

The FPGA-Based Traffic Monitoring and Signal Control System project demonstrates a compact digital-logic design mapped to an FPGA. The expected output table provides a reference for checking the RTL in simulation and on the board. For a complete submission, include the actual Questa waveform, transcript, Quartus compilation evidence, and video link in the corresponding repository folders.

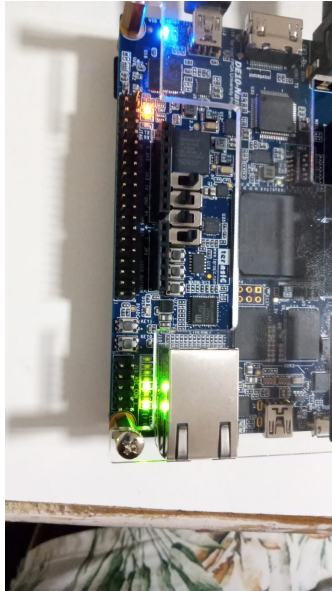
## Project 4 — Intermediate — Hardware Photo Evidence



FPGA-Based Traffic Monitoring and Signal Control System hardware evidence — photo 1



FPGA-Based Traffic Monitoring and Signal Control System hardware evidence — photo 2



FPGA-Based Traffic Monitoring and Signal Control System hardware evidence — photo 3

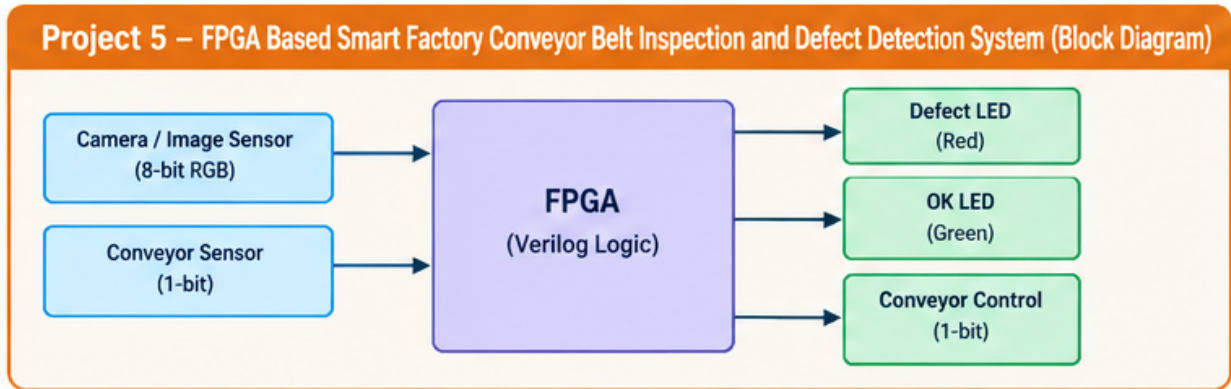
# Project 5 — Advanced

## FPGA-Based Conveyor Belt Inspection System

### 1. Objective

Detect an object and classify it as normal or defective from a 4-bit sensor pattern, activating an alarm for defective patterns.

### 2. Block Diagram



FPGA-Based Conveyor Belt Inspection System — block diagram

### 3. RTL Design

**Inputs:** sensor[3:0]

**Outputs:** object\_detected, normal\_led, defective\_led, alarm\_led

**Logic summary:** object\_detected = sensor[0]. A normal object is the exact pattern sensor[3:0] = 0111. Other detected patterns meeting the defective condition assert defective\_led and alarm\_led.

```
module Conveyor_Inspection (
    input wire [3:0] sensor,
    output wire object_detected,
    output wire normal_led,
    output wire defective_led,
    output wire alarm_led
);
    assign object_detected = sensor[0];
    assign normal_led =
        sensor[0] &
        sensor[1] &
        sensor[2] &
        ~sensor[3];
    assign defective_led =
        sensor[0] &
        (~sensor[1] | ~sensor[2] | sensor[3]);
    assign alarm_led = defective_led;
endmodule
```

### 4. Simulation Results

| Input case | Expected outputs   | Interpretation       |
|------------|--------------------|----------------------|
| 0111       | Object=1, Normal=1 | Defective=0, Alarm=0 |
| 0000       | Object=0, Normal=0 | Defective=0, Alarm=0 |
| 0001       | Object=1, Normal=0 | Defective=1, Alarm=1 |
| 1111       | Object=1, Normal=0 | Defective=1, Alarm=1 |

### 5. Hardware Implementation

The RTL is intended to be compiled for the DE10-Nano Cyclone V FPGA and assigned to onboard switches and LEDs using the project pin assignments. The photos below document the powered board state supplied for this project. LED patterns in photos can be affected by switch positions, camera exposure, and viewing orientation.

## Pin assignment summary

| Signal          | Assignment                    |
|-----------------|-------------------------------|
| sensor[0..3]    | SW0..SW3 / Y24, W24, W21, W20 |
| object_detected | LED0 / W15                    |
| normal_led      | LED1 / AA24                   |
| defective_led   | LED2 / V16                    |
| alarm_led       | LED3 / V15                    |

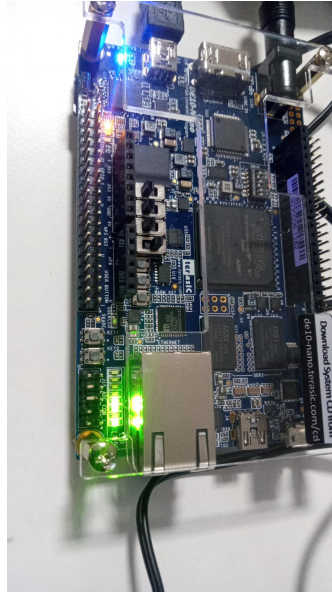
## 6. Applications

Educational prototype for digital inspection logic, object-present sensing, defect flagging, and conveyor quality-control concepts.

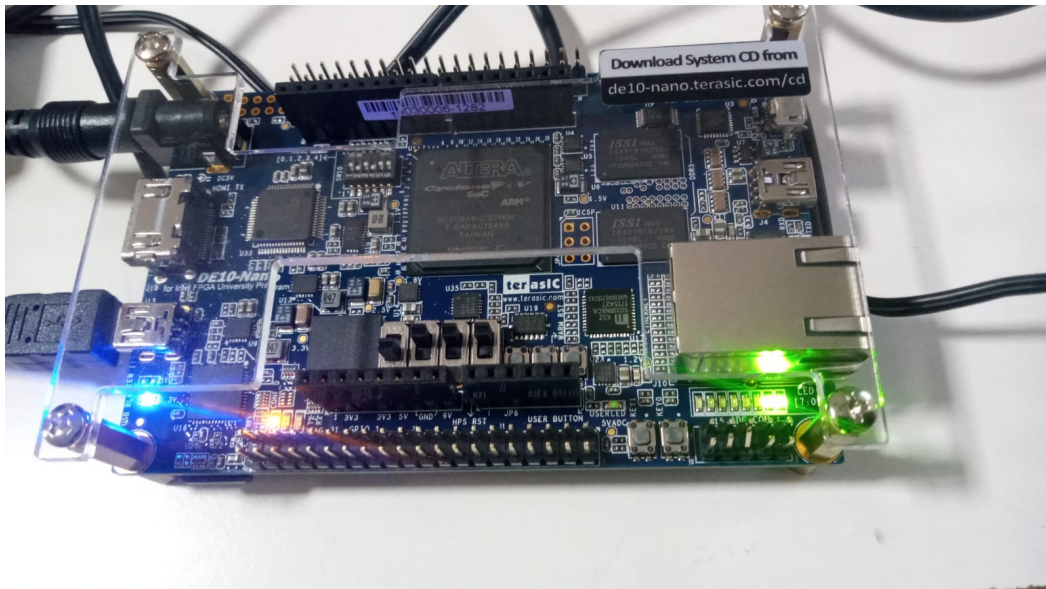
## 7. Conclusion

The FPGA-Based Conveyor Belt Inspection System project demonstrates a compact digital-logic design mapped to an FPGA. The expected output table provides a reference for checking the RTL in simulation and on the board. For a complete submission, include the actual Questa waveform, transcript, Quartus compilation evidence, and video link in the corresponding repository folders.

## Project 5 — Advanced — Hardware Photo Evidence



FPGA-Based Conveyor Belt Inspection System hardware evidence — photo 1



FPGA-Based Conveyor Belt Inspection System hardware evidence — photo 2

## Final Submission Checklist

- Documentation: one overall PDF plus individual project README files.
- RTL: top-level Verilog file and any submodules, with module names matching the testbenches.
- Testbench: one testbench per project with boundary cases and representative inputs.
- Simulation: waveform image, transcript, and simulation report PDF per project.
- Images: block diagram, RTL schematic, board setup photo, hardware output photo, and useful pin/compile screenshots.
- Video: working Google Drive link in each Video\_Link.txt, with permissions set so evaluators can view it.
- Reproducibility: verify device selection, top-level entity, pin assignments, compilation status, and board behavior before final upload.

Prepared for the V-SPACE FPGA Build Challenge 2026 submission. Confirm all paths, screenshots, and repository links before uploading the final package.